# Solucionario: representación de la información y circuitos lógicos

Documento para el profesorado. Resultados finales de cada apartado; el alumnado debe mostrar el procedimiento.

## Ejercicio 1

**1.** $133{,}3_6$ = **57,5** · $10110000{,}111_2$ = **176,875**

**2.** a) 214,1875 = **$11010110{,}0011_2$** = **$326{,}14_8$** = **$D6{,}3_{16}$**. b) $110{,}000110_2$ (6 bits fraccionarios), no es exacto: se almacena **6,09375**; error = **0,00625**.

**3.** $6CF{,}B_{16}$ = $11011001111{,}1011_2$ = **$3317{,}54_8$** · $2531{,}5_8$ = $10101011001{,}101_2$ = **$559{,}A_{16}$**

**4.** a) **1000110,1** (51,75 + 18,75 = 70,5) · b) **11010** (40,75 − 14,75 = 26)

**5.** a) SM **10111001**, C1 **11000110**, C2 **11000111**. b) 00010100 + 10101101 = **11000001** = -63. c) 00011000 + 00101001 = 01000001 → se lee 65. **No hay overflow**, el resultado es correcto. d) Signos distintos: 83 − 20 = 63, gana el positivo → **00111111**.

**6.** 3,25 = 0011,0100; 6 = 0110,0000 → C2 = 1010,0000. Suma = **1101,0100** = −8·1 + ... = **−2,75**

**7.** a) **1011010** (10·9 = 90). b) **1000000,11** (9,25·7 = 64,75). c) ≪2: **1000111000** = 568 (×4). ≫3: **00010001** = 17 (división entera entre 8; se pierden los bits 110, resto 6).

**8.** 3,2 GB · 8 = 25,6 Gb = 25.600 Mb → **42,67 s**. Tamaño: **2,98 GiB**.

**9.** S = (A + B)’ + B ⊕ C. Intermedias: (A + B)’ y B ⊕ C. S = 1 en ABC = **000, 001, 010, 101, 110**.

| A | B | C | (A + B)’ | B ⊕ C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **1** |
| 0 | 0 | 1 | 1 | 1 | **1** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 0 | **0** |
| 1 | 0 | 1 | 0 | 1 | **1** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A’BC + AB’C’ + ABC. Simplificada: **S = AB’C’ + BC**

## Ejercicio 2

**1.** $736{,}6_8$ = **478,75** · $10100001{,}101_2$ = **161,625**

**2.** a) 153,1875 = **$10011001{,}0011_2$** = **$231{,}14_8$** = **$99{,}3_{16}$**. b) $11100{,}010011_2$ (6 bits fraccionarios), no es exacto: se almacena **28,296875**; error = **0,003125**.

**3.** $25D{,}8_{16}$ = $1001011101{,}1_2$ = **$1135{,}4_8$** · $5347{,}6_8$ = $101011100111{,}11_2$ = **$AE7{,}C_{16}$**

**4.** a) **1001001,11** (33,5 + 40,25 = 73,75) · b) **1000100,01** (98,5 − 30,25 = 68,25)

**5.** a) SM **11000111**, C1 **10111000**, C2 **10111001**. b) 01000110 + 10010010 = **11011000** = -40. c) 10101110 + 10011100 = 01001010 (acarreo descartado) → se lee 74. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-182 no cabe en −128…127). d) Signos distintos: 110 − 70 = 40, gana el positivo → **00101000**.

**6.** 1,5 = 0001,1000; 3,75 = 0011,1100 → C2 = 1100,0100. Suma = **1101,1100** = −8·1 + … = **−2,25**

**7.** a) **1010001** (27·3 = 81). b) **111101,01** (8,75·7 = 61,25). c) ≪2: **1100001000** = 776 (×4). ≫3: **00011000** = 24 (división entera entre 8; se pierden los bits 010, resto 2).

**8.** 6.220.800 bytes = **6,221 MB** = **5,933 MiB**. Caben **5144** imágenes.

**9.** S = ((A + B)’ + C’)’. Intermedias: (A + B)’ y C’. S = 1 en ABC = **011, 101, 111**.

| A | B | C | (A + B)’ | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **0** |
| 0 | 0 | 1 | 1 | 0 | **0** |
| 0 | 1 | 0 | 0 | 1 | **0** |
| 0 | 1 | 1 | 0 | 0 | **1** |
| 1 | 0 | 0 | 0 | 1 | **0** |
| 1 | 0 | 1 | 0 | 0 | **1** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A’B’C’ + A’B’C + A’BC + AB’C. Simplificada: **S = A’B’ + A’C + B’C**

## Ejercicio 3

**1.** $47B,4B_{16}$ = **1147,29296875** · $1100100,101_2$ = **100,625**

**2.** a) 187,1875 = **$10111011,0011_2$** = **$273,14_8$** = **$BB,3_{16}$**. b) $11100,010011_2$ (6 bits fraccionarios), no es exacto: se almacena **28,296875**; error = **0,003125**.

**3.** $2AA,F_{16}$ = $1010101010,1111_2$ = **$1252,74_8$** · $5141,3_8$ = $101001100001,011_2$ = **$A61,6_{16}$**

**4.** a) **110101,01** (33,5 + 19,75 = 53,25) · b) **1011101,011** (112,25 − 18,875 = 93,375)

**5.** a) SM **10011001**, C1 **11100110**, C2 **11100111**. b) 00111110 + 10110100 = **11110010** = -14. c) 00100110 + 00101010 = 01010000 → se lee 80. **No hay overflow**, el resultado es correcto. d) Signos distintos: 76 − 62 = 14, gana el positivo → **00001110**.

**6.** 1,25 = 0001,0100; 1,75 = 0001,1100 → C2 = 1110,0100. Suma = **1111,1000** = −8·1 + ... = **−0,5**

**7.** a) **101000101** (25·13 = 325). b) **100000,01** (10,75·3 = 32,25). c) ≪2: **1001110100** = 628 (×4). ≫3: **00010011** = 19 (división entera entre 8; se pierden los bits 101, resto 5).

**8.** 8.589.934.592 bytes = 68.719.476.736 bits = **8,59 GB**. 8.589.934.592 direcciones → **33 bits** ($2^{33}$).

**9.** S = A·B ⊕ (A + C). Intermedias: A·B y A + C. S = 1 en ABC = **001, 011, 100, 101**.

| A | B | C | A·B | A + C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | 1 | **1** |
| 0 | 1 | 0 | 0 | 0 | **0** |
| 0 | 1 | 1 | 0 | 1 | **1** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 1 | **1** |
| 1 | 1 | 0 | 1 | 1 | **0** |
| 1 | 1 | 1 | 1 | 1 | **0** |

**10.** Canónica: A’B’C’ + A’BC + AB’C’ + AB’C. Simplificada: **S = A’BC + AB’ + B’C’**

## Ejercicio 4

**1.** $311{,}3_5$ = **81,6** · $11110000{,}011_2$ = **240,375**

**2.** a) 179,6875 = **$10110011{,}1011_2$** = **$263{,}54_8$** = **$B3{,}B_{16}$**. b) $10101{,}010011_2$ (6 bits fraccionarios), no es exacto: se almacena **21,296875**; error = **0,003125**.

**3.** $58B{,}8_{16}$ = $10110001011{,}1_2$ = **$2613{,}4_8$** · $4540{,}3_8$ = $100101100000{,}011_2$ = **$960{,}6_{16}$**

**4.** a) **1101111,11** (89,25 + 22,5 = 111,75) · b) **11110,11** (42,25 − 11,5 = 30,75)

**5.** a) SM **10011010**, C1 **11100101**, C2 **11100110**. b) 00010011 + 10001111 = **10100010** = -94. c) 01011101 + 01101000 = 11000101 → se lee -59. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (197 no cabe en −128…127). d) Signos distintos: 113 − 19 = 94, gana el positivo → **01011110**.

**6.** 4 = 0100,0000; 6 = 0110,0000 → C2 = 1010,0000. Suma = **1110,0000** = −8·1 + … = **−2**

**7.** a) **11001000** (20·10 = 200). b) **1011001,01** (12,75·7 = 89,25). c) ≪2: **1101101100** = 876 (×4). ≫3: **00011011** = 27 (división entera entre 8; se pierden los bits 011, resto 3).

**8.** a) 48 kB = **384.000 bits**; 48 KiB = **393.216 bits**. b) 134.217.728 bytes = **134,218 MB**. Kibi/Mebi usan 1024; kilo/mega, 1000.

**9.** S = A ⊕ B ⊕ C’. Intermedias: A ⊕ B y C’. S = 1 en ABC = **000, 011, 101, 110**.

| A | B | C | A ⊕ B | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **0** |
| 0 | 1 | 1 | 1 | 0 | **1** |
| 1 | 0 | 0 | 1 | 1 | **0** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: AB’C + ABC’ + ABC. Simplificada: **S = AB + AC**

## Ejercicio 5

**1.** $245{,}3_{6}$ = **101,5** · $1001111{,}001_{2}$ = **79,125**

**2.** a) 228,5625 = **$11100100{,}1001_{2}$** = **$344{,}44_{8}$** = **$E4{,}9_{16}$**. b) $10111{,}100110_{2}$ (6 bits fraccionarios), no es exacto: se almacena **23,59375**; error = **0,00625**.

**3.** $9A9{,}D_{16}$ = $100110101001{,}1101_{2}$ = **$4651{,}64_{8}$** · $1706{,}2_{8}$ = $1111000110{,}01_{2}$ = **$3C6{,}4_{16}$**

**4.** a) **1011011,1** (33,25 + 58,25 = 91,5) · b) **1001011,011** (85,5 − 10,125 = 75,375)

**5.** a) SM **11001000**, C1 **10110111**, C2 **10111000**. b) 01101010 + 10010101 = **11111111** = -1. c) 00101100 + 00011100 = 01001000 → se lee 72. **No hay overflow**, el resultado es correcto. d) Signos distintos: 107 − 106 = 1, gana el positivo → **00000001**.

**6.** 2,25 = 0010,0100; 3,25 = 0011,0100 → C2 = 1100,1100. Suma = **1111,0000** = −8·1 + ... = **−1**

**7.** a) **10000010** (26·5 = 130). b) **1001001,11** (14,75·5 = 73,75). c) ≪2: **1011010100** = 724 (×4). ≫3: **00010110** = 22 (división entera entre 8; se pierden los bits 101, resto 5).

**8.** 500.000.000.000 bytes / $2^{30}$ = **465,66 «GB»**; diferencia ≈ **6,87 %**. Vídeos: **310**.

**9.** S = (A + B)’ + C’. Intermedias: (A + B)’ y C’. S = 1 en ABC = **000, 001, 010, 100, 110**.

| A | B | C | (A + B)’ | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **1** |
| 0 | 0 | 1 | 1 | 0 | **1** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A’B’C + A’BC’ + A’BC + AB’C’ + AB’C. Simplificada: **S = A’B + A’C + AB’**

## Ejercicio 6

**1.** $543{,}4_8$ = **355,5** · $10101000{,}001_2$ = **168,125**

**2.** a) 170,6875 = **$10101010{,}1011_2$** = **$252{,}54_8$** = **$AA{,}B_{16}$**. b) $10100{,}100110_2$ (6 bits fraccionarios), no es exacto: se almacena **20,59375**; error = **0,00625**.

**3.** $9B5{,}4_{16}$ = $100110110101{,}01_2$ = **$4665{,}2_8$** · $3410{,}4_8$ = $11100001000{,}1_2$ = **$708{,}8_{16}$**

**4.** a) **1000100** (50,25 + 17,75 = 68) · b) **111,001** (44,5 − 37,375 = 7,125)

**5.** a) SM **11110000**, C1 **10001111**, C2 **10010000**. b) 00010001 + 10001110 = **10011111** = -97. c) 11000000 + 10101001 = 01101001 (acarreo descartado) → se lee 105. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-151 no cabe en −128…127). d) Signos distintos: 114 − 17 = 97, gana el positivo → **01100001**.

**6.** 1,75 = 0001,1100; 6,5 = 0110,1000 → C2 = 1001,1000. Suma = **1011,0100** = −8·1 + … = **−4,75**

**7.** a) **110110** (18·3 = 54). b) **10010,11** (6,25·3 = 18,75). c) ≪2: **0110111000** = 440 (×4). ≫3: **00001101** = 13 (división entera entre 8; se pierden los bits 110, resto 6).

**8.** 4.000.000.000.000 bytes; Windows: **3725,29 «GB»** (GiB) = **3,638 TiB**; df -H: 4,0T.

**9.** S = (A + B)·C’. Intermedias: A + B y C’. S = 1 en ABC = **010, 100, 110**.

| A | B | C | A + B | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **0** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **1** |
| 0 | 1 | 1 | 1 | 0 | **0** |
| 1 | 0 | 0 | 1 | 1 | **1** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 1 | 1 | **1** |
| 1 | 1 | 1 | 1 | 0 | **0** |

**10.** Canónica: A’B’C’ + A’BC’ + A’BC + AB’C’. Simplificada: **S = A’B + B’C’**

## Ejercicio 7

**1.** $72B,C_{16}$ = **1835,75** · $11101110,111_2$ = **238,875**

**2.** a) 144,6875 = **$10010000,1011_2$** = **$220,54_8$** = **$90,B_{16}$**. b) $10011,011001_2$ (6 bits fraccionarios), no es exacto: se almacena **19,390625**; error = **0,009375**.

**3.** $490,E_{16}$ = $10010010000,111_2$ = **$2220,7_8$** · $5233,4_8$ = $101010011011,1_2$ = **$A9B,8_{16}$**

**4.** a) **10010101,01** (88,5 + 60,75 = 149,25) · b) **101011,101** (72,75 − 29,125 = 43,625)

**5.** a) SM **11011100**, C1 **10100011**, C2 **10100100**. b) 00101000 + 10001010 = **10110010** = -78. c) 11010101 + 11001111 = 10100100 (acarreo descartado) → se lee -92. **No hay overflow**, el resultado es correcto. d) Signos distintos: 118 − 40 = 78, gana el positivo → **01001110**.

**6.** 1,75 = 0001,1100; 3,75 = 0011,1100 → C2 = 1100,0100. Suma = **1110,0000** = −8·1 + ... = **−2**

**7.** a) **100100** (12·3 = 36). b) **100010,1** (5,75·6 = 34,5). c) ≪2: **0101110100** = 372 (×4). ≫3: **00001011** = 11 (división entera entre 8; se pierden los bits 101, resto 5).

**8.** 8,5 GB · 8 = 68 Gb = 68.000 Mb → **68 s**. Tamaño: **7,916 GiB**.

**9.** S = ((A·B)'·(A ⊕ C))'. Intermedias: (A·B)' y A ⊕ C. S = 1 en ABC = **000, 010, 101, 110, 111**.

| A | B | C | (A·B)' | A ⊕ C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **1** |
| 0 | 0 | 1 | 1 | 1 | **0** |
| 0 | 1 | 0 | 1 | 0 | **1** |
| 0 | 1 | 1 | 1 | 1 | **0** |
| 1 | 0 | 0 | 1 | 1 | **0** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A'B'C' + A'BC + AB'C'. Simplificada: **S = A'BC + B'C'**

## Ejercicio 8

**1.** $320{,}3_5$ = **85,6** · $1001010{,}001_2$ = **74,125**

**2.** a) 218,6875 = **$11011010{,}1011_2$** = **$332{,}54_8$** = **$DA{,}B_{16}$**. b) $10011{,}110011_2$ (6 bits fraccionarios), no es exacto: se almacena **19,796875**; error = **0,003125**.

**3.** $D89{,}B_{16}$ = $110110001001{,}1011_2$ = **$6611{,}54_8$** · $1056{,}5_8$ = $1000101110{,}101_2$ = **$22E{,}A_{16}$**

**4.** a) **1011110,101** (33,875 + 60,75 = 94,625) · b) **100010,001** (58,5 − 24,375 = 34,125)

**5.** a) SM **10010110**, C1 **11101001**, C2 **11101010**. b) 00110110 + 10110001 = **11100111** = -25. c) 01011100 + 01001000 = 10100100 → se lee -92. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (164 no cabe en −128…127). d) Signos distintos: 79 − 54 = 25, gana el positivo → **00011001**.

**6.** 2,5 = 0010,1000; 4,75 = 0100,1100 → C2 = 1011,0100. Suma = **1101,1100** = −8·1 + … = **−2,25**

**7.** a) **101010010** (26·13 = 338). b) **100011,01** (11,75·3 = 35,25). c) ≪2: **0110000100** = 388 (×4). ≫3: **00001100** = 12 (división entera entre 8; se pierden los bits 001, resto 1).

**8.** 11.059.200 bytes = **11,059 MB** = **10,547 MiB**. Caben **2893** imágenes.

**9.** S = (A·B)’·(A ⊕ C). Intermedias: (A·B)’ y A ⊕ C. S = 1 en ABC = **001, 011, 100**.

| A | B | C | (A·B)’ | A ⊕ C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **0** |
| 0 | 0 | 1 | 1 | 1 | **1** |
| 0 | 1 | 0 | 1 | 0 | **0** |
| 0 | 1 | 1 | 1 | 1 | **1** |
| 1 | 0 | 0 | 1 | 1 | **1** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A’B’C’ + A’BC’ + A’BC. Simplificada: **S = A’B + A’C’**

## Ejercicio 9

**1.** $424{,}3_{6}$ = **160,5** · $10011011{,}001_{2}$ = **155,125**

**2.** a) 161,8125 = **$10100001{,}1101_{2}$** = **$241{,}64_{8}$** = **$A1{,}D_{16}$**. b) $10100{,}100110_{2}$ (6 bits fraccionarios), no es exacto: se almacena **20,59375**; error = **0,00625**.

**3.** $BE2{,}9_{16}$ = $101111100010{,}1001_{2}$ = **$5742{,}44_{8}$** · $765{,}7_{8}$ = $111110101{,}111_{2}$ = **$1F5{,}E_{16}$**

**4.** a) **1101010,001** (87,625 + 18,5 = 106,125) · b) **1001000,011** (88,75 − 16,375 = 72,375)

**5.** a) SM **10111110**, C1 **11000001**, C2 **11000010**. b) 01110000 + 10001011 = **11111011** = -5. c) 11010000 + 11001010 = 10011010 (acarreo descartado) → se lee -102. **No hay overflow**, el resultado es correcto. d) Signos distintos: 117 − 112 = 5, gana el positivo → **00000101**.

**6.** 0,75 = 0000,1100; 3,25 = 0011,0100 → C2 = 1100,1100. Suma = **1101,1000** = −8·1 + ... = **−2,5**

**7.** a) **100011110** (22·13 = 286). b) **11111,01** (6,25·5 = 31,25). c) ≪2: **0010100000** = 160 (×4). ≫3: **00000101** = 5 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** 34.359.738.368 bytes = 274.877.906.944 bits = **34,36 GB**. 34.359.738.368 direcciones → **35 bits** ($2^{35}$).

**9.** S = (A ⊕ B)·(A·C)'. Intermedias: A ⊕ B y (A·C)'. S = 1 en ABC = **010, 011, 100**.

| A | B | C | A ⊕ B | (A·C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **0** |
| 0 | 0 | 1 | 0 | 1 | **0** |
| 0 | 1 | 0 | 1 | 1 | **1** |
| 0 | 1 | 1 | 1 | 1 | **1** |
| 1 | 0 | 0 | 1 | 1 | **1** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A'BC' + AB'C' + AB'C + ABC' + ABC. Simplificada: **S = A + BC'**

## Ejercicio 10

**1.** $612{,}5_8$ = **394,625** · $10000010{,}001_2$ = **130,125**

**2.** a) 179,8125 = **$10110011{,}1101_2$** = **$263{,}64_8$** = **$B3{,}D_{16}$**. b) $1111{,}001100_2$ (6 bits fraccionarios), no es exacto: se almacena **15,1875**; error = **0,0125**.

**3.** $D5A{,}8_{16}$ = $110101011010{,}1_2$ = **$6532{,}4_8$** · $1320{,}6_8$ = $1011010000{,}11_2$ = **$2D0{,}C_{16}$**

**4.** a) **1010111,111** (27,625 + 60,25 = 87,875) · b) **111010,111** (81,5 − 22,625 = 58,875)

**5.** a) SM **11010110**, C1 **10101001**, C2 **10101010**. b) 00011111 + 10010110 = **10110101** = -75. c) 10110011 + 10010100 = 01000111 (acarreo descartado) → se lee 71. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-185 no cabe en −128…127). d) Signos distintos: 106 − 31 = 75, gana el positivo → **01001011**.

**6.** 0,25 = 0000,0100; 2,5 = 0010,1000 → C2 = 1101,1000. Suma = **1101,1100** = −8·1 + … = **−2,25**

**7.** a) **1101100** (12·9 = 108). b) **101111,01** (6,75·7 = 47,25). c) ≪2: **0100010000** = 272 (×4). ≫3: **00001000** = 8 (división entera entre 8; se pierden los bits 100, resto 4).

**8.** a) 96 kB = **768.000 bits**; 96 KiB = **786.432 bits**. b) 134.217.728 bytes = **134,218 MB**. Kibi/Mebi usan 1024; kilo/mega, 1000.

**9.** S = A·B·(B + C). Intermedias: A·B y B + C. S = 1 en ABC = **110, 111**.

| A | B | C | A·B | B + C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | 1 | **0** |
| 0 | 1 | 0 | 0 | 1 | **0** |
| 0 | 1 | 1 | 0 | 1 | **0** |
| 1 | 0 | 0 | 0 | 0 | **0** |
| 1 | 0 | 1 | 0 | 1 | **0** |
| 1 | 1 | 0 | 1 | 1 | **1** |
| 1 | 1 | 1 | 1 | 1 | **1** |

**10.** Canónica: A’B’C’ + A’BC’ + AB’C’ + ABC. Simplificada: **S = A’C’ + ABC + B’C’**

## Ejercicio 11

**1.** $26D,CE_{16}$ = **621,8046875** · $11101101,011_2$ = **237,375**

**2.** a) 179,1875 = **$10110011,0011_2$** = **$263,14_8$** = **$B3,3_{16}$**. b) $10110,001100_2$ (6 bits fraccionarios), no es exacto: se almacena **22,1875**; error = **0,0125**.

**3.** $8EA,5_{16}$ = $100011101010,0101_2$ = **$4352,24_8$** · $2004,6_8$ = $10000000100,11_2$ = **$404,C_{16}$**

**4.** a) **1011111,01** (39,75 + 55,5 = 95,25) · b) **1010000,1** (112,25 − 31,75 = 80,5)

**5.** a) SM **11000010**, C1 **10111101**, C2 **10111110**. b) 01010110 + 10010101 = **11101011** = -21. c) 11011001 + 11011110 = 10110111 (acarreo descartado) → se lee -73. **No hay overflow**, el resultado es correcto. d) Signos distintos: 107 − 86 = 21, gana el positivo → **00010101**.

**6.** 0,5 = 0000,1000; 5,5 = 0101,1000 → C2 = 1010,1000. Suma = **1011,0000** = −8·1 + ... = **−5**

**7.** a) **1111000** (20·6 = 120). b) **110111,1** (9,25·6 = 55,5). c) ≪2: **0111101100** = 492 (×4). ≫3: **00001111** = 15 (división entera entre 8; se pierden los bits 011, resto 3).

**8.** 750.000.000.000 bytes / $2^{30}$ = **698,49 «GB»**; diferencia ≈ **6,87 %**. Vídeos: **465**.

**9.** S = (A·B)’ ⊕ B·C. Intermedias: (A·B)’ y B·C. S = 1 en ABC = **000, 001, 010, 100, 101, 111**.

| A | B | C | (A·B)’ | B·C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **1** |
| 0 | 0 | 1 | 1 | 0 | **1** |
| 0 | 1 | 0 | 1 | 0 | **1** |
| 0 | 1 | 1 | 1 | 1 | **0** |
| 1 | 0 | 0 | 1 | 0 | **1** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 0 | **0** |
| 1 | 1 | 1 | 0 | 1 | **1** |

**10.** Canónica: A’B’C’ + A’BC’ + ABC. Simplificada: **S = A’C’ + ABC**

## Ejercicio 12

**1.** $422{,}2_5$ = **112,4** · $10100110{,}111_2$ = **166,875**

**2.** a) 203,1875 = **$11001011{,}0011_2$** = **$313{,}14_8$** = **$CB{,}3_{16}$**. b) $11011{,}111001_2$ (6 bits fraccionarios), no es exacto: se almacena **27,890625**; error = **0,009375**.

**3.** $9E1{,}F_{16}$ = $100111100001{,}1111_2$ = **$4741{,}74_8$** · $6170{,}3_8$ = $110001111000{,}011_2$ = **$C78{,}6_{16}$**

**4.** a) **1110000** (86,5 + 25,5 = 112) · b) **110100,01** (85,75 − 33,5 = 52,25)

**5.** a) SM **10011101**, C1 **11100010**, C2 **11100011**. b) 00111110 + 10110011 = **11110001** = -15. c) 01010010 + 01100111 = 10111001 → se lee -71. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (185 no cabe en −128…127). d) Signos distintos: 77 − 62 = 15, gana el positivo → **00001111**.

**6.** 1 = 0001,0000; 3,75 = 0011,1100 → C2 = 1100,0100. Suma = **1101,0100** = −8·1 + … = **−2,75**

**7.** a) **110110** (9·6 = 54). b) **1001001,1** (12,25·6 = 73,5). c) ≪2: **1001110100** = 628 (×4). ≫3: **00010011** = 19 (división entera entre 8; se pierden los bits 101, resto 5).

**8.** 2.000.000.000.000 bytes; Windows: **1862,65 «GB»** (GiB) = **1,819 TiB**; df -H: 2,0T.

**9.** S = (A ⊕ B)·(A + C). Intermedias: A ⊕ B y A + C. S = 1 en ABC = **011, 100, 101**.

| A | B | C | A ⊕ B | A + C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | 1 | **0** |
| 0 | 1 | 0 | 1 | 0 | **0** |
| 0 | 1 | 1 | 1 | 1 | **1** |
| 1 | 0 | 0 | 1 | 1 | **1** |
| 1 | 0 | 1 | 1 | 1 | **1** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 1 | **0** |

**10.** Canónica: A’B’C’ + A’BC’ + A’BC + ABC. Simplificada: **S = A’C’ + BC**

## Ejercicio 13

**1.** $104{,}3_6$ = **40,5** · $10101001{,}011_2$ = **169,375**

**2.** a) 163,1875 = **$10100011{,}0011_{\mathbf{2}}$** = **$243{,}14_{\mathbf{8}}$** = **$A3{,}3_{\mathbf{16}}$**. b) $10000{,}010011_2$ (6 bits fraccionarios), no es exacto: se almacena **16,296875**; error = **0,003125**.

**3.** $1BD{,}6_{16}$ = $110111101{,}011_2$ = **$675{,}3_{\mathbf{8}}$** · $4465{,}1_8$ = $100100110101{,}001_2$ = **$935{,}2_{\mathbf{16}}$**

**4.** a) **1101110,001** (84,375 + 25,75 = 110,125) · b) **110101,011** (73,5 − 20,125 = 53,375)

**5.** a) SM **11010111**, C1 **10101000**, C2 **10101001**. b) 00010110 + 11011100 = **11110010** = -14. c) 11100010 + 11001101 = 10101111 (acarreo descartado) → se lee -81. **No hay overflow**, el resultado es correcto. d) Signos distintos: 36 − 22 = 14, gana el positivo → **00001110**.

**6.** 0,5 = 0000,1000; 1,5 = 0001,1000 → C2 = 1110,1000. Suma = **1111,0000** = −8·1 + ... = **−1**

**7.** a) **1101100** (18·6 = 108). b) **100001,11** (11,25·3 = 33,75). c) ≪2: **1100001100** = 780 (×4). ≫3: **00011000** = 24 (división entera entre 8; se pierden los bits 011, resto 3).

**8.** 8,5 GB · 8 = 68 Gb = 68.000 Mb → **113,33 s**. Tamaño: **7,916 GiB**.

**9.** S = A ⊕ B ⊕ (A + C). Intermedias: A ⊕ B y A + C. S = 1 en ABC = **001, 010, 110, 111**.

| A | B | C | A ⊕ B | A + C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | 1 | **1** |
| 0 | 1 | 0 | 1 | 0 | **1** |
| 0 | 1 | 1 | 1 | 1 | **0** |
| 1 | 0 | 0 | 1 | 1 | **0** |
| 1 | 0 | 1 | 1 | 1 | **0** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 1 | **1** |

**10.** Canónica: A’B’C + A’BC’ + A’BC + AB’C’ + ABC. Simplificada: **S = A’B + A’C + AB’C’ + BC**

## Ejercicio 14

**1.** $247{,}4_8$ = **167,5** · $10000000{,}011_2$ = **128,375**

**2.** a) 142,8125 = **$10001110{,}1101_2$** = **$216{,}64_8$** = **$8E{,}D_{16}$**. b) $110{,}010011_2$ (6 bits fraccionarios), no es exacto: se almacena **6,296875**; error = **0,003125**.

**3.** $A37{,}A_{16}$ = $101000110111{,}101_2$ = **$5067{,}5_8$** · $3662{,}5_8$ = $11110110010{,}101_2$ = **$7B2{,}A_{16}$**

**4.** a) **1010000,101** (58,375 + 22,25 = 80,625) · b) **1000000,001** (102,25 − 38,125 = 64,125)

**5.** a) SM **11000001**, C1 **10111110**, C2 **10111111**. b) 00010001 + 10001001 = **10011010** = -102. c) 01001011 + 01101010 = 10110101 → se lee -75. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (181 no cabe en −128…127). d) Signos distintos: 119 − 17 = 102, gana el positivo → **01100110**.

**6.** 0,75 = 0000,1100; 3,25 = 0011,0100 → C2 = 1100,1100. Suma = **1101,1000** = −8·1 + … = **−2,5**

**7.** a) **1010100** (12·7 = 84). b) **100110,11** (7,75·5 = 38,75). c) ≪2: **0011110000** = 240 (×4). ≫3: **00000111** = 7 (división entera entre 8; se pierden los bits 100, resto 4).

**8.** 14.745.600 bytes = **14,746 MB** = **14,062 MiB**. Caben **2170** imágenes.

**9.** S = A·B + C’. Intermedias: A·B y C’. S = 1 en ABC = **000, 010, 100, 110, 111**.

| A | B | C | A·B | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 1 | 1 | **1** |
| 1 | 1 | 1 | 1 | 0 | **1** |

**10.** Canónica: A’B’C’ + A’BC + ABC’ + ABC. Simplificada: **S = A’B’C’ + AB + BC**

## Ejercicio 15

**1.** $68D,CB_{16}$ = **1677,79296875** · $11001101,011_2$ = **205,375**

**2.** a) 141,3125 = **$10001101,0101_2$** = **$215,24_8$** = **$8D,5_{16}$**. b) $11001,010011_2$ (6 bits fraccionarios), no es exacto: se almacena **25,296875**; error = **0,003125**.

**3.** $A39,2_{16}$ = $101000111001,001_2$ = **$5071,1_8$** · $2137,2_8$ = $10001011111,01_2$ = **$45F,4_{16}$**

**4.** a) **101001,011** (30,625 + 10,75 = 41,375) · b) **100101** (64,75 − 27,75 = 37)

**5.** a) SM **11100101**, C1 **10011010**, C2 **10011011**. b) 00100111 + 10100111 = **11001110** = -50. c) 11011101 + 11000111 = 10100100 (acarreo descartado) → se lee -92. **No hay overflow**, el resultado es correcto. d) Signos distintos: 89 − 39 = 50, gana el positivo → **00110010**.

**6.** 3 = 0011,0000; 6,25 = 0110,0100 → C2 = 1001,1100. Suma = **1100,1100** = −8·1 + ... = **−3,25**

**7.** a) **100010011** (25·11 = 275). b) **111000,01** (11,25·5 = 56,25). c) ≪2: **0010100100** = 164 (×4). ≫3: **00000101** = 5 (división entera entre 8; se pierden los bits 001, resto 1).

**8.** 34.359.738.368 bytes = 274.877.906.944 bits = **34,36 GB**. 34.359.738.368 direcciones → **35 bits** ($2^{35}$).

**9.** S = (A·B)'·(A + C)'. Intermedias: (A·B)' y (A + C)'. S = 1 en ABC = **000, 010**.

| A | B | C | (A·B)' | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **1** |
| 0 | 0 | 1 | 1 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **1** |
| 0 | 1 | 1 | 1 | 0 | **0** |
| 1 | 0 | 0 | 1 | 0 | **0** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 0 | 0 | **0** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A'B'C' + AB'C + ABC' + ABC. Simplificada: **S = A'B'C' + AB + AC**

## Ejercicio 16

**1.** $242{,}2_5$ = **72,4** · $1111111{,}001_2$ = **127,125**

**2.** a) 174,1875 = **$10101110{,}0011_2$** = **$256{,}14_8$** = **$AE{,}3_{16}$**. b) $10111{,}111001_2$ (6 bits fraccionarios), no es exacto: se almacena **23,890625**; error = **0,009375**.

**3.** $B1A{,}3_{16}$ = $101100011010{,}0011_2$ = **$5432{,}14_8$** · $4017{,}4_8$ = $100000001111{,}1_2$ = **$80F{,}8_{16}$**

**4.** a) **1001000,011** (45,625 + 26,75 = 72,375) · b) **110011,11** (76,25 − 24,5 = 51,75)

**5.** a) SM **10100010**, C1 **11011101**, C2 **11011110**. b) 00111000 + 11000000 = **11111000** = -8. c) 11000000 + 10111001 = 01111001 (acarreo descartado) → se lee 121. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-135 no cabe en −128…127). d) Signos distintos: 64 − 56 = 8, gana el positivo → **00001000**.

**6.** 5 = 0101,0000; 5,75 = 0101,1100 → C2 = 1010,0100. Suma = **1111,0100** = −8·1 + … = **−0,75**

**7.** a) **11110010** (22·11 = 242). b) **10101,11** (7,25·3 = 21,75). c) ≪2: **0101010000** = 336 (×4). ≫3: **00001010** = 10 (división entera entre 8; se pierden los bits 100, resto 4).

**8.** a) 96 kB = **768.000 bits**; 96 KiB = **786.432 bits**. b) 134.217.728 bytes = **134,218 MB**. Kibi/Mebi usan 1024; kilo/mega, 1000.

**9.** S = (A·B + (A + C)')'. Intermedias: A·B y (A + C)'. S = 1 en ABC = **001, 011, 100, 101**.

| A | B | C | A·B | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **0** |
| 0 | 0 | 1 | 0 | 0 | **1** |
| 0 | 1 | 0 | 0 | 1 | **0** |
| 0 | 1 | 1 | 0 | 0 | **1** |
| 1 | 0 | 0 | 0 | 0 | **1** |
| 1 | 0 | 1 | 0 | 0 | **1** |
| 1 | 1 | 0 | 1 | 0 | **0** |
| 1 | 1 | 1 | 1 | 0 | **0** |

**10.** Canónica: A'B'C' + A'B'C + A'BC' + AB'C' + AB'C. Simplificada: **S = A'C' + B'**

## Ejercicio 17

**1.** $522{,}3_{6}$ = **194,5** · $10110011{,}001_{2}$ = **179,125**

**2.** a) 150,8125 = **$10010110{,}1101_{2}$** = **$226{,}64_{8}$** = **$96{,}D_{16}$**. b) $11001{,}000110_{2}$ (6 bits fraccionarios), no es exacto: se almacena **25,09375**; error = **0,00625**.

**3.** $36E{,}4_{16}$ = $1101101110{,}01_{2}$ = **$1556{,}2_{8}$** · $2575{,}6_{8}$ = $10101111101{,}11_{2}$ = **$57D{,}C_{16}$**

**4.** a) **1110111,11** (71,5 + 48,25 = 119,75) · b) **1000111,011** (84,25 − 12,875 = 71,375)

**5.** a) SM **10110100**, C1 **11001011**, C2 **11001100**. b) 00011111 + 11001001 = **11101000** = -24. c) 00101011 + 00101111 = 01011010 → se lee 90. **No hay overflow**, el resultado es correcto. d) Signos distintos: 55 − 31 = 24, gana el positivo → **00011000**.

**6.** 3,5 = 0011,1000; 4 = 0100,0000 → C2 = 1100,0000. Suma = **1111,1000** = −8·1 + … = **−0,5**

**7.** a) **1111000** (24·5 = 120). b) **100100,11** (12,25·3 = 36,75). c) ≪2: **1010001100** = 652 (×4). ≫3: **00010100** = 20 (división entera entre 8; se pierden los bits 011, resto 3).

**8.** 500.000.000.000 bytes / $2^{30}$ = **465,66 «GB»**; diferencia ≈ **6,87 %**. Vídeos: **310**.

**9.** S = (A + B) ⊕ C'. Intermedias: A + B y C'. S = 1 en ABC = **000, 011, 101, 111**.

| A | B | C | A + B | C' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **0** |
| 0 | 1 | 1 | 1 | 0 | **1** |
| 1 | 0 | 0 | 1 | 1 | **0** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 1 | 1 | **0** |
| 1 | 1 | 1 | 1 | 0 | **1** |

**10.** Canónica: A'B'C' + A'BC + AB'C + ABC'. Simplificada: **S = A'B'C' + A'BC + AB'C + ABC'**

## Ejercicio 18

**1.** $534{,}7_8$ = **348,875** · $11101101{,}011_2$ = **237,375**

**2.** a) 133,6875 = **$10000101{,}1011_2$** = **$205{,}54_8$** = **$85{,}B_{16}$**. b) $10000{,}100110_2$ (6 bits fraccionarios), no es exacto: se almacena **16,59375**; error = **0,00625**.

**3.** $17E{,}5_{16}$ = $101111110{,}0101_2$ = **$576{,}24_8$** · $3040{,}1_8$ = $11000100000{,}001_2$ = **$620{,}2_{16}$**

**4.** a) **1110000,111** (89,375 + 23,5 = 112,875) · b) **1000110,01** (105,75 − 35,5 = 70,25)

**5.** a) SM **11001001**, C1 **10110110**, C2 **10110111**. b) 00101110 + 10101011 = **11011001** = -39. c) 01100010 + 01010011 = 10110101 → se lee -75. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (181 no cabe en −128…127). d) Signos distintos: 85 − 46 = 39, gana el positivo → **00100111**.

**6.** 2 = 0010,0000; 6,5 = 0110,1000 → C2 = 1001,1000. Suma = **1011,1000** = −8·1 + … = **−4,5**

**7.** a) **10001010** (23·6 = 138). b) **100011,01** (11,75·3 = 35,25). c) ≪2: **1010000000** = 640 (×4). ≫3: **00010100** = 20 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** 8.000.000.000.000 bytes; Windows: **7450,58 «GB»** (GiB) = **7,276 TiB**; df -H: 8,0T.

**9.** S = ((A·B)'·(A + C)')'. Intermedias: (A·B)' y (A + C)'. S = 1 en ABC = **001, 011, 100, 101, 110, 111**.

| A | B | C | (A·B)' | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **0** |
| 0 | 0 | 1 | 1 | 0 | **1** |
| 0 | 1 | 0 | 1 | 1 | **0** |
| 0 | 1 | 1 | 1 | 0 | **1** |
| 1 | 0 | 0 | 1 | 0 | **1** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 0 | **1** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A'B'C + AB'C' + AB'C. Simplificada: **S = AB' + B'C**

## Ejercicio 19

**1.** $E7F,C1_{16}$ = **3711,75390625** · $10011000,011_2$ = **152,375**

**2.** a) 240,4375 = **$11110000,0111_2$** = **$360,34_8$** = **$F0,7_{16}$**. b) $110,100110_2$ (6 bits fraccionarios), no es exacto: se almacena **6,59375**; error = **0,00625**.

**3.** $96D,1_{16}$ = $100101101101,0001_2$ = **$4555,04_8$** · $2535,7_8$ = $10101011101,111_2$ = **$55D,E_{16}$**

**4.** a) **110011** (40,75 + 10,25 = 51) · b) **101101,01** (84,5 − 39,25 = 45,25)

**5.** a) SM **11011000**, C1 **10100111**, C2 **10101000**. b) 00010100 + 11011011 = **11101111** = -17. c) 11010011 + 11010111 = 10101010 (acarreo descartado) → se lee -86. **No hay overflow**, el resultado es correcto. d) Signos distintos: 37 − 20 = 17, gana el positivo → **00010001**.

**6.** 5 = 0101,0000; 6,5 = 0110,1000 → C2 = 1001,1000. Suma = **1110,1000** = $-8 \cdot 1 + \ldots$ = **−1,5**

**7.** a) **1010000** ($16 \cdot 5 = 80$). b) **111010,11** (11,75·5 = 58,75). c) ≪2: **1100000100** = 772 (×4). ≫3: **00011000** = 24 (división entera entre 8; se pierden los bits 001, resto 1).

**8.** 4,7 GB · 8 = 37,6 Gb = 37.600 Mb → **47 s**. Tamaño: **4,377 GiB**.

**9.** S = (A + B)’ + B ⊕ C. Intermedias: (A + B)’ y B ⊕ C. S = 1 en ABC = **000, 001, 010, 101, 110**.

| A | B | C | (A + B)’ | B ⊕ C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **1** |
| 0 | 0 | 1 | 1 | 1 | **1** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 0 | **0** |
| 1 | 0 | 1 | 0 | 1 | **1** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A’B’C’ + A’B’C + A’BC’ + ABC. Simplificada: **S = A’B’ + A’C’ + ABC**

## Ejercicio 20

**1.** $432{,}2_5$ = **117,4** · $11100001{,}011_2$ = **225,375**

**2.** a) 137,6875 = **$10001001{,}1011_2$** = **$211{,}54_8$** = **$89{,}B_{16}$**. b) $11011{,}000110_2$ (6 bits fraccionarios), no es exacto: se almacena **27,09375**; error = **0,00625**.

**3.** $94E{,}9_{16}$ = $100101001110{,}1001_2$ = **$4516{,}44_8$** · $2553{,}1_8$ = $10101101011{,}001_2$ = **$56B{,}2_{16}$**

**4.** a) **1100101,101** (84,875 + 16,75 = 101,625) · b) **11011,011** (62,25 − 34,875 = 27,375)

**5.** a) SM **10100100**, C1 **11011011**, C2 **11011100**. b) 01000010 + 10100111 = **11101001** = -23. c) 10101110 + 10011010 = 01001000 (acarreo descartado) → se lee 72. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-184 no cabe en −128…127). d) Signos distintos: 89 − 66 = 23, gana el positivo → **00010111**.

**6.** 1,5 = 0001,1000; 4,75 = 0100,1100 → C2 = 1011,0100. Suma = **1100,1100** = −8·1 + … = **−3,25**

**7.** a) **10000010** (26·5 = 130). b) **100110,01** (12,75·3 = 38,25). c) ≪2: **0110010100** = 404 (×4). ≫3: **00001100** = 12 (división entera entre 8; se pierden los bits 101, resto 5).

**8.** 6.220.800 bytes = **6,221 MB** = **5,933 MiB**. Caben **5144** imágenes.

**9.** S = A ⊕ B ⊕ (B + C)'. Intermedias: A ⊕ B y (B + C)'. S = 1 en ABC = **000, 010, 011, 101**.

| A | B | C | A ⊕ B | (B + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 0 | **1** |
| 0 | 1 | 1 | 1 | 0 | **1** |
| 1 | 0 | 0 | 1 | 1 | **0** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 0 | 0 | **0** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A'B'C' + A'B'C + A'BC + AB'C + ABC'. Simplificada: **S = A'B' + A'C + ABC' + B'C**

## Ejercicio 21

**1.** $510{,}3_6$ = **186,5** · $10011100{,}011_2$ = **156,375**

**2.** a) 241,1875 = **$11110001{,}0011_2$** = **$361{,}14_8$** = **$F1{,}3_{16}$**. b) $1000{,}011001_2$ (6 bits fraccionarios), no es exacto: se almacena **8,390625**; error = **0,009375**.

**3.** $1CD{,}8_{16}$ = $111001101{,}1_2$ = **$715{,}4_8$** · $4412{,}3_8$ = $100100001010{,}011_2$ = **$90A{,}6_{16}$**

**4.** a) **1110100,101** (88,125 + 28,5 = 116,625) · b) **101111,101** (66,5 − 18,875 = 47,625)

**5.** a) SM **11110000**, C1 **10001111**, C2 **10010000**. b) 00010101 + 10001111 = **10100100** = -92. c) 11100110 + 11001011 = 10110001 (acarreo descartado) → se lee -79. **No hay overflow**, el resultado es correcto. d) Signos distintos: 113 − 21 = 92, gana el positivo → **01011100**.

**6.** 4,75 = 0100,1100; 5,75 = 0101,1100 → C2 = 1010,0100. Suma = **1111,0000** = −8·1 + ... = **−1**

**7.** a) **11010000** (16·13 = 208). b) **111111,11** (12,75·5 = 63,75). c) ≪2: **1001001000** = 584 (×4). ≫3: **00010010** = 18 (división entera entre 8; se pierden los bits 010, resto 2).

**8.** 34.359.738.368 bytes = 274.877.906.944 bits = **34,36 GB**. 34.359.738.368 direcciones → **35 bits** ($2^{35}$).

**9.** S = (A + B)’ + C’. Intermedias: (A + B)’ y C’. S = 1 en ABC = **000, 001, 010, 100, 110**.

| A | B | C | (A + B)’ | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **1** |
| 0 | 0 | 1 | 1 | 0 | **1** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 0 | 1 | **1** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A’BC + AB’C’ + ABC. Simplificada: **S = AB’C’ + BC**

## Ejercicio 22

**1.** $715{,}2_8$ = **461,25** · $10011110{,}111_2$ = **158,875**

**2.** a) 159,4375 = **$10011111{,}0111_2$** = **$237{,}34_8$** = **$9F{,}7_{16}$**. b) $101{,}001100_2$ (6 bits fraccionarios), no es exacto: se almacena **5,1875**; error = **0,0125**.

**3.** $2FE{,}F_{16}$ = $1011111110{,}1111_2$ = **$1376{,}74_8$** · $6477{,}5_8$ = $110100111111{,}101_2$ = **$D3F{,}A_{16}$**

**4.** a) **10010000,1** (86,75 + 57,75 = 144,5) · b) **110110,101** (76,75 − 22,125 = 54,625)

**5.** a) SM **11100101**, C1 **10011010**, C2 **10011011**. b) 00110101 + 10101101 = **11100010** = -30. c) 01001010 + 01101101 = 10110111 → se lee -73. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (183 no cabe en −128…127). d) Signos distintos: 83 − 53 = 30, gana el positivo → **00011110**.

**6.** 0,75 = 0000,1100; 3,25 = 0011,0100 → C2 = 1100,1100. Suma = **1101,1000** = −8·1 + … = **−2,5**

**7.** a) **100101001** (27·11 = 297). b) **10101,11** (7,25·3 = 21,75). c) ≪2: **1101000000** = 832 (×4). ≫3: **00011010** = 26 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** a) 48 kB = **384.000 bits**; 48 KiB = **393.216 bits**. b) 536.870.912 bytes = **536,871 MB**. Kibi/Mebi usan 1024; kilo/mega, 1000.

**9.** S = ((A + B)’ + A ⊕ C)’. Intermedias: (A + B)’ y A ⊕ C. S = 1 en ABC = **010, 101, 111**.

| A | B | C | (A + B)’ | A ⊕ C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **0** |
| 0 | 0 | 1 | 1 | 1 | **0** |
| 0 | 1 | 0 | 0 | 0 | **1** |
| 0 | 1 | 1 | 0 | 1 | **0** |
| 1 | 0 | 0 | 0 | 1 | **0** |
| 1 | 0 | 1 | 0 | 0 | **1** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A’B’C’ + A’BC’ + AB’C + ABC’ + ABC. Simplificada: **S = A’C’ + AB + AC**

## Ejercicio 23

**1.** $79E,B8_{16}$ = **1950,71875** · $11101111,101_2$ = **239,625**

**2.** a) 173,1875 = **$10101101,0011_2$** = **$255,14_8$** = **$AD,3_{16}$**. b) $10101,100110_2$ (6 bits fraccionarios), no es exacto: se almacena **21,59375**; error = **0,00625**.

**3.** $92A,2_{16}$ = $100100101010,001_2$ = **$4452,1_8$** · $6515,4_8$ = $110101001101,1_2$ = **$D4D,8_{16}$**

**4.** a) **1010000,111** (20,375 + 60,5 = 80,875) · b) **111000,101** (81,75 − 25,125 = 56,625)

**5.** a) SM **10011001**, C1 **11100110**, C2 **11100111**. b) 00010010 + 10111010 = **11001100** = -52. c) 00111001 + 00101011 = 01100100 → se lee 100. **No hay overflow**, el resultado es correcto. d) Signos distintos: 70 − 18 = 52, gana el positivo → **00110100**.

**6.** 1,25 = 0001,0100; 2,25 = 0010,0100 → C2 = 1101,1100. Suma = **1111,0000** = −8·1 + ... = **−1**

**7.** a) **11000110** (22·9 = 198). b) **11100,11** (5,75·5 = 28,75). c) ≪2: **1011011000** = 728 (×4). ≫3: **00010110** = 22 (división entera entre 8; se pierden los bits 110, resto 6).

**8.** 960.000.000.000 bytes / $2^{30}$ = **894,07 «GB»**; diferencia ≈ **6,87 %**. Vídeos: **596**.

**9.** S = (A + B)’ + B·C. Intermedias: (A + B)’ y B·C. S = 1 en ABC = **000, 001, 011, 111**.

| A | B | C | (A + B)’ | B·C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | **1** |
| 0 | 0 | 1 | 1 | 0 | **1** |
| 0 | 1 | 0 | 0 | 0 | **0** |
| 0 | 1 | 1 | 0 | 1 | **1** |
| 1 | 0 | 0 | 0 | 0 | **0** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 0 | 0 | **0** |
| 1 | 1 | 1 | 0 | 1 | **1** |

**10.** Canónica: A’B’C + AB’C + ABC’. Simplificada: **S = ABC’ + B’C**

## Ejercicio 24

**1.** $402{,}4_5$ = **102,8** · $1000001{,}011_2$ = **65,375**

**2.** a) 239,8125 = **$11101111{,}1101_2$** = **$357{,}64_8$** = **$EF{,}D_{16}$**. b) $10000{,}011001_2$ (6 bits fraccionarios), no es exacto: se almacena **16,390625**; error = **0,009375**.

**3.** $BE6{,}3_{16}$ = $101111100110{,}0011_2$ = **$5746{,}14_8$** · $5514{,}5_8$ = $1011010011 00{,}101_2$ = **$B4C{,}A_{16}$**

**4.** a) **1111101,11** (72,25 + 53,5 = 125,75) · b) **1001100,1** (115,25 − 38,75 = 76,5)

**5.** a) SM **11101011**, C1 **10010100**, C2 **10010101**. b) 00010101 + 10111100 = **11010001** = -47. c) 01101001 + 01101001 = 11010010 → se lee -46. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (210 no cabe en −128…127). d) Signos distintos: 68 − 21 = 47, gana el positivo → **00101111**.

**6.** 3 = 0011,0000; 5,75 = 0101,1100 → C2 = 1010,0100. Suma = **1101,0100** = −8·1 + … = **−2,75**

**7.** a) **10010000** (16·9 = 144). b) **1010101,1** (14,25·6 = 85,5). c) ≪2: **1001101100** = 620 (×4). ≫3: **00010011** = 19 (división entera entre 8; se pierden los bits 011, resto 3).

**8.** 8.000.000.000.000 bytes; Windows: **7450,58 «GB»** (GiB) = **7,276 TiB**; df -H: 8,0T.

**9.** S = A·B + C’. Intermedias: A·B y C’. S = 1 en ABC = **000, 010, 100, 110, 111**.

| A | B | C | A·B | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 1 | 1 | **1** |
| 1 | 1 | 1 | 1 | 0 | **1** |

**10.** Canónica: A’B’C + A’BC’ + AB’C’ + AB’C + ABC. Simplificada: **S = A’BC’ + AB’ + AC + B’C**

## Ejercicio 25

**1.** $201{,}3_{6}$ = **73,5** · $1101001{,}001_{2}$ = **105,125**

**2.** a) 218,8125 = **$11011010{,}1101_{2}$** = **$332{,}64_{8}$** = **$DA{,}D_{16}$**. b) $111{,}110011_{2}$ (6 bits fraccionarios), no es exacto: se almacena **7,796875**; error = **0,003125**.

**3.** $7CD{,}C_{16}$ = $11111001101{,}11_{2}$ = **$3715{,}6_{8}$** · $723{,}5_{8}$ = $111010011{,}101_{2}$ = **$1D3{,}A_{16}$**

**4.** a) **10000111,001** (74,625 + 60,5 = 135,125) · b) **1001111** (108,5 − 29,5 = 79)

**5.** a) SM **10111101**, C1 **11000010**, C2 **11000011**. b) 00010010 + 11010010 = **11100100** = -28. c) 00011000 + 00010100 = 00101100 → se lee 44. **No hay overflow**, el resultado es correcto. d) Signos distintos: 46 − 18 = 28, gana el positivo → **00011100**.

**6.** 1,75 = 0001,1100; 5 = 0101,0000 → C2 = 1011,0000. Suma = **1100,1100** = −8·1 + ... = **−3,25**

**7.** a) **10110110** (26·7 = 182). b) **1000000,11** (9,25·7 = 64,75). c) ≪2: **1100100000** = 800 (×4). ≫3: **00011001** = 25 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** 4,7 GB · 8 = 37,6 Gb = 37.600 Mb → **62,67 s**. Tamaño: **4,377 GiB**.

**9.** S = (A + B) ⊕ (A + C)'. Intermedias: A + B y (A + C)'. S = 1 en ABC = **000, 011, 100, 101, 110, 111**.

| A | B | C | A + B | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **0** |
| 0 | 1 | 1 | 1 | 0 | **1** |
| 1 | 0 | 0 | 1 | 0 | **1** |
| 1 | 0 | 1 | 1 | 0 | **1** |
| 1 | 1 | 0 | 1 | 0 | **1** |
| 1 | 1 | 1 | 1 | 0 | **1** |

**10.** Canónica: A'BC + AB'C + ABC. Simplificada: **S = AC + BC**

## Ejercicio 26

**1.** $701{,}2_8$ = **449,25** · $1011000{,}101_2$ = **88,625**

**2.** a) 226,4375 = **$11100010{,}0111_2$** = **$342{,}34_8$** = **$E2{,}7_{16}$**. b) $1110{,}001100_2$ (6 bits fraccionarios), no es exacto: se almacena **14,1875**; error = **0,0125**.

**3.** $2B5{,}3_{16}$ = $1010110101{,}0011_2$ = **$1265{,}14_8$** · $2361{,}2_8$ = $10011110001{,}01_2$ = **$4F1{,}4_{16}$**

**4.** a) **1101001,1** (69,75 + 35,75 = 105,5) · b) **111000,01** (85,5 − 29,25 = 56,25)

**5.** a) SM **11010111**, C1 **10101000**, C2 **10101001**. b) 00110010 + 10001001 = **10111011** = -69. c) 10100001 + 10110111 = 01011000 (acarreo descartado) → se lee 88. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-168 no cabe en −128…127). d) Signos distintos: 119 − 50 = 69, gana el positivo → **01000101**.

**6.** 4,25 = 0100,0100; 5 = 0101,0000 → C2 = 1011,0000. Suma = **1111,0100** = −8·1 + … = **−0,75**

**7.** a) **100011110** (26·11 = 286). b) **1011000,1** (14,75·6 = 88,5). c) ≪2: **1011000000** = 704 (×4). ≫3: **00010110** = 22 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** 11.059.200 bytes = **11,059 MB** = **10,547 MiB**. Caben **2893** imágenes.

**9.** S = ((A + B)’ + C’)’. Intermedias: (A + B)’ y C’. S = 1 en ABC = **011, 101, 111**.

| A | B | C | (A + B)’ | C’ | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **0** |
| 0 | 0 | 1 | 1 | 0 | **0** |
| 0 | 1 | 0 | 0 | 1 | **0** |
| 0 | 1 | 1 | 0 | 0 | **1** |
| 1 | 0 | 0 | 0 | 1 | **0** |
| 1 | 0 | 1 | 0 | 0 | **1** |
| 1 | 1 | 0 | 0 | 1 | **0** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A’BC + AB’C’ + ABC’. Simplificada: **S = A’BC + AC’**

## Ejercicio 27

**1.** $2DE,6A_{16}$ = **734,4140625** · $10111111,101_2$ = **191,625**

**2.** a) 209,6875 = **$11010001,1011_2$** = **$321,54_8$** = **$D1,B_{16}$**. b) $1011,010011_2$ (6 bits fraccionarios), no es exacto: se almacena **11,296875**; error = **0,003125**.

**3.** $ABE,D_{16}$ = $101010111110,1101_2$ = **$5276,64_8$** · $2026,6_8$ = $10000010110,11_2$ = **$416,C_{16}$**

**4.** a) **111101,11** (26,5 + 35,25 = 61,75) · b) **1101000,1** (119,25 − 14,75 = 104,5)

**5.** a) SM **10110111**, C1 **11001000**, C2 **11001001**. b) 00010110 + 10010111 = **10101101** = -83. c) 00011111 + 00101011 = 01001010 → se lee 74. **No hay overflow**, el resultado es correcto. d) Signos distintos: 105 − 22 = 83, gana el positivo → **01010011**.

**6.** 1,25 = 0001,0100; 4,25 = 0100,0100 → C2 = 1011,1100. Suma = **1101,0000** = −8·1 + ... = **−3**

**7.** a) **110010** (10·5 = 50). b) **100111,11** (13,25·3 = 39,75). c) ≪2: **0110000100** = 388 (×4). ≫3: **00001100** = 12 (división entera entre 8; se pierden los bits 001, resto 1).

**8.** 17.179.869.184 bytes = 137.438.953.472 bits = **17,18 GB**. 17.179.869.184 direcciones → **34 bits** ($2^{34}$).

**9.** S = (A + B)·B·C. Intermedias: A + B y B·C. S = 1 en ABC = **011, 111**.

| A | B | C | A + B | B·C | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | **0** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | 0 | **0** |
| 0 | 1 | 1 | 1 | 1 | **1** |
| 1 | 0 | 0 | 1 | 0 | **0** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 1 | 0 | **0** |
| 1 | 1 | 1 | 1 | 1 | **1** |

**10.** Canónica: A’B’C’ + A’B’C + A’BC’ + AB’C. Simplificada: **S = A’C’ + B’C**

## Ejercicio 28

**1.** $112{,}4_5$ = **32,8** · $1000011{,}011_2$ = **67,375**

**2.** a) 200,1875 = **$11001000{,}0011_2$** = **$310{,}14_8$** = **$C8{,}3_{16}$**. b) $10001{,}110011_2$ (6 bits fraccionarios), no es exacto: se almacena **17,796875**; error = **0,003125**.

**3.** $264{,}A_{16}$ = $1001100100{,}101_2$ = **$1144{,}5_8$** · $475{,}6_8$ = $100111101{,}11_2$ = **$13D{,}C_{16}$**

**4.** a) **10000000,001** (74,875 + 53,25 = 128,125) · b) **1001111** (102,25 − 23,25 = 79)

**5.** a) SM **10101110**, C1 **11010001**, C2 **11010010**. b) 00100010 + 11011101 = **11111111** = -1. c) 10100001 + 10110000 = 01010001 (acarreo descartado) → se lee 81. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (-175 no cabe en −128…127). d) Signos distintos: 35 − 34 = 1, gana el positivo → **00000001**.

**6.** 0,75 = 0000,1100; 3,5 = 0011,1000 → C2 = 1100,1000. Suma = **1101,0100** = −8·1 + … = **−2,75**

**7.** a) **10001111** (13·11 = 143). b) **1010101,1** (14,25·6 = 85,5). c) ≪2: **0100000000** = 256 (×4). ≫3: **00001000** = 8 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** a) 48 kB = **384.000 bits**; 48 KiB = **393.216 bits**. b) 134.217.728 bytes = **134,218 MB**. Kibi/Mebi usan 1024; kilo/mega, 1000.

**9.** S = A·B ⊕ C'. Intermedias: A·B y C'. S = 1 en ABC = **000, 010, 100, 111**.

| A | B | C | A·B | C' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **1** |
| 0 | 0 | 1 | 0 | 0 | **0** |
| 0 | 1 | 0 | 0 | 1 | **1** |
| 0 | 1 | 1 | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | 1 | **1** |
| 1 | 0 | 1 | 0 | 0 | **0** |
| 1 | 1 | 0 | 1 | 1 | **0** |
| 1 | 1 | 1 | 1 | 0 | **1** |

**10.** Canónica: A'B'C' + A'BC' + A'BC + AB'C' + ABC. Simplificada: **S = A'B + B'C' + BC**

## Ejercicio 29

**1.** $223{,}3_{6}$ = **87,5** · $11010110{,}001_{2}$ = **214,125**

**2.** a) 133,1875 = **$10000101{,}0011_{2}$** = **$205{,}14_{8}$** = **$85{,}3_{16}$**. b) $1011{,}001100_{2}$ (6 bits fraccionarios), no es exacto: se almacena **11,1875**; error = **0,0125**.

**3.** $5A0{,}7_{16}$ = $10110100000{,}0111_{2}$ = **$2640{,}34_{8}$** · $3153{,}2_{8}$ = $11001101011{,}01_{2}$ = **$66B{,}4_{16}$**

**4.** a) **1100110,101** (78,125 + 24,5 = 102,625) · b) **11001,001** (46,5 − 21,375 = 25,125)

**5.** a) SM **10111110**, C1 **11000001**, C2 **11000010**. b) 00100101 + 11001111 = **11110100** = -12. c) 11010110 + 11010110 = 10101100 (acarreo descartado) → se lee -84. **No hay overflow**, el resultado es correcto. d) Signos distintos: 49 − 37 = 12, gana el positivo → **00001100**.

**6.** 0,25 = 0000,0100; 5 = 0101,0000 → C2 = 1011,0000. Suma = **1011,0100** = −8·1 + ... = **−4,75**

**7.** a) **10000010** (26·5 = 130). b) **1000100,11** (13,75·5 = 68,75). c) ≪2: **0110100000** = 416 (×4). ≫3: **00001101** = 13 (división entera entre 8; se pierden los bits 000, resto 0).

**8.** 750.000.000.000 bytes / $2^{30}$ = **698,49 «GB»**; diferencia ≈ **6,87 %**. Vídeos: **465**.

**9.** S = (A·B)'·(A + C)'. Intermedias: (A·B)' y (A + C)'. S = 1 en ABC = **000, 010**.

| A | B | C | (A·B)' | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | **1** |
| 0 | 0 | 1 | 1 | 0 | **0** |
| 0 | 1 | 0 | 1 | 1 | **1** |
| 0 | 1 | 1 | 1 | 0 | **0** |
| 1 | 0 | 0 | 1 | 0 | **0** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 0 | 0 | **0** |
| 1 | 1 | 1 | 0 | 0 | **0** |

**10.** Canónica: A'B'C + AB'C' + ABC. Simplificada: **S = A'B'C + AB'C' + ABC**

## Ejercicio 30

**1.** $172{,}2_8$ = **122,25** · $10111110{,}111_2$ = **190,875**

**2.** a) 151,5625 = $\mathbf{10010111{,}1001_2}$ = $\mathbf{227{,}44_8}$ = $\mathbf{97{,}9_{16}}$. b) $1010{,}000110_2$ (6 bits fraccionarios), no es exacto: se almacena **10,09375**; error = **0,00625**.

**3.** $AF0{,}E_{16}$ = $101011110000{,}111_2$ = $\mathbf{5360{,}7_8}$ · $1222{,}3_8$ = $1010010010{,}011_2$ = $\mathbf{292{,}6_{16}}$

**4.** a) **1101000,101** (87,375 + 17,25 = 104,625) · b) **1000011,101** (93,25 − 25,625 = 67,625)

**5.** a) SM **11010010**, C1 **10101101**, C2 **10101110**. b) 00111100 + 10110001 = **11101101** = -19. c) 01100000 + 01101011 = 11001011 → se lee -53. **Overflow**: dos operandos del mismo signo dan resultado de signo contrario (203 no cabe en −128…127). d) Signos distintos: 79 − 60 = 19, gana el positivo → **00010011**.

**6.** 4,75 = 0100,1100; 6,25 = 0110,0100 → C2 = 1001,1100. Suma = **1110,1000** = −8·1 + … = **−1,5**

**7.** a) **10011100** (26·6 = 156). b) **11111,1** (5,25·6 = 31,5). c) ≪2: **1001101000** = 616 (×4). ≫3: **00010011** = 19 (división entera entre 8; se pierden los bits 010, resto 2).

**8.** 4.000.000.000.000 bytes; Windows: **3725,29 «GB»** (GiB) = **3,638 TiB**; df -H: 4,0T.

**9.** S = (A ⊕ B + (A + C)')'. Intermedias: A ⊕ B y (A + C)'. S = 1 en ABC = **001, 110, 111**.

| A | B | C | A ⊕ B | (A + C)' | S |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | **0** |
| 0 | 0 | 1 | 0 | 0 | **1** |
| 0 | 1 | 0 | 1 | 1 | **0** |
| 0 | 1 | 1 | 1 | 0 | **0** |
| 1 | 0 | 0 | 1 | 0 | **0** |
| 1 | 0 | 1 | 1 | 0 | **0** |
| 1 | 1 | 0 | 0 | 0 | **1** |
| 1 | 1 | 1 | 0 | 0 | **1** |

**10.** Canónica: A'B'C + A'BC' + A'BC + ABC'. Simplificada: **S = A'C + BC'**